Generalisation example 1: Wisconsin breast cancer (numeric, binary, no test file)

Matric No.: <MATRIC NO> **Name:** <FULL NAME> **Generalisation example** **Variant-2**

Model: `Claude Opus 5.5 (claude-opus-5-5)` | Interface: Agent harness - Claude Code 2.1.278 (desktop app) | Code: `https://github.com/BreezyB1n/in6227-assignment1`

Summary. Same skill and code as the main report, a very different dataset: 30 numeric cell-morphology features, no categorical columns, no missing values and no supplied test file (a stratified 20% hold-out is created). LR and Random Forest are statistically indistinguishable on the test set (ROC-AUC difference +0.001, 95% CI -0.004 to +0.008).

1 Data exploration and cleaning

455 training and 114 test rows; 37.4% malignant, i.e. *not* imbalanced by the skill's 35% rule, so the primary metric switches to ROC-AUC and the threshold objective to balanced accuracy. The profiler raised 18 flags; the new detectors behaved sensibly (the drift detector, which now requires KS $p < 0.01$, no longer fires on the 114-row hold-out).

Table 1: Data issues found by the profiler and the treatment chosen (all statistics learned inside CV folds).

Issue	Evidence	Treatment and rationale
No test file	stratified 20% hold-out (seed 42)	Hold-out created once with a fixed seed and used once.
Heavy tails	7 *error* columns with skew ≥ 2 (max skew 5.50)	LR: `log1p` (values are non-negative); RF: none (rank-based splits).
Co-occurring zeros	6 concavity columns zero together in 11 rows, all benign	Kept as values: a convex contour *has* concavity 0, so these are real measurements, not sentinels.
Near-exact redundancy	`mean perimeter` rebuilt from other columns with R^2=0.9997	LR: drop `mean perimeter` (perimeter $\approx 2\pi r$); RF: keep (A2).
Correlated size measures	0.998 (radius, perimeter) and many more	Keep; L2 copes. Importances are shared.

2 Feature selection and engineering

The same family-specific logic as in the main report: transform and de-duplicate for the linear model only, leave the forest alone. The concavity zeros are the instructive contrast with the main dataset: the detector fires on both, but here the decision is the opposite (keep), because the zeros have a physical meaning.

3 Model training

LR (one-hot is a no-op; standardised) vs. **Random Forest instead of gradient boosting**: the house rule for $n < 1000$ applies, because early stopping would hold out only ∼45 rows. The forest has no stopping rule (300 trees fixed; more trees do not overfit), so depth, leaf size and `max_features` are tuned. 10-fold CV (rule for $n < 2000$). The selected RF `min_samples_leaf=1` lies on its range boundary, but 1 is the smallest legal value, so the range cannot be widened (logged). LR converged after 35 iterations.

Table 2: Hyper-parameter search (randomised, stratified CV, refit on the primary metric) and fitted models. Gap = CV train − validation primary metric (overfitting indicator).

Model	Search	Selected	Stopping	#inputs	Gap
Logistic Regression	20 cand. over C [0.001, 100] log, class_weight ∈ {none, balanced}	C=1.14, class_weight=None	35/5000 iter.	29	+0.002
Random Forest	20 cand. over max_depth ∈ {none, 4, 8, 16}, min_samples_leaf [1, 20], max_features ∈ {sqrt, 0.3, 0.6}, class_weight ∈ {none, balanced}	class_weight=balanced, max_depth=16, max_features=sqrt, min_samples_leaf=1 (boundary: min_samples_leaf)	–	30	+0.011

4 Evaluation and comparison

Table 3: Cross-validated and held-out test performance. Precision/recall/F1 are for the positive class at threshold t, tuned on out-of-fold training predictions; AP = average precision (PR-AUC). CIs: 1000 test-set bootstrap resamples.

Model	CV ROC-AUC (mean±sd)	Test AP [95% CI]	ROC-AUC	t	Prec.	Recall	F1	Macro-F1	Bal. Acc.
Prior baseline	–	0.368	0.500	–	–	0.000	0.000	0.387	0.500
Logistic Regression	0.996±0.006	0.994 [0.979, 1.000]	0.996	0.41	0.976	0.952	0.964	0.972	0.969
Random Forest	0.989±0.016	0.995 [0.985, 1.000]	0.997	0.46	1.000	0.929	0.963	0.971	0.964

Both models are near the ceiling. The paired comparison finds no difference: McNemar counts 2/2 (p=1.000), CV corrected t-test p=0.235. With 114 test rows, one misclassification moves recall by 2.4 points, so the CIs are wide.

Table 4: Ablations: one decision reversed, tuned hyper-parameters kept (so Δ is conservative for the variant). Δ = change in CV roc_auc on the same folds; sd = sd of paired fold differences ($|\Delta| <$ sd: no detectable effect); p: Nadeau–Bengio corrected t-test.

ID	Question	Δ roc_auc	sd	p
A1_lr_no_log	LR: no log transform of skewed error columns	-0.0010	0.0022	0.342
A2_lr_keep_perimeter	LR: keep the redundant mean perimeter	-0.0004	0.0009	0.329

5 Findings and discussion

On a small, clean, numeric dataset the skill correctly chose different rules (ROC-AUC, 10 folds, Random Forest, keep zeros) and concluded *no detectable difference*, preferring LR for its simplicity. Neither LR preprocessing step (log transform, dropping the redundant perimeter) changed CV ROC-AUC beyond fold noise.

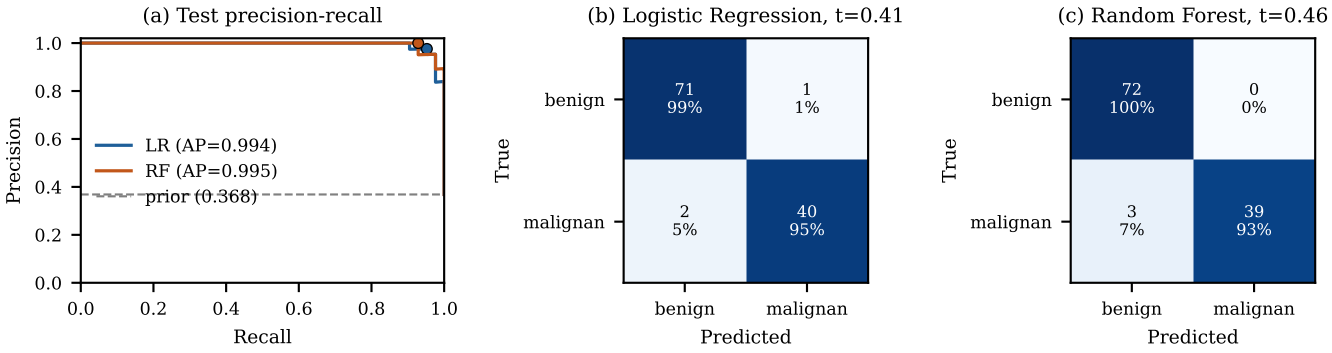

Figure 1: Hold-out set. (a) Precision–recall curves with the tuned operating points. (b, c) Confusion matrices at the tuned thresholds.